

CELTIC KNOT

MATHEMATICS

A Coloring Book of Interlace & Weave

50 Knotwork Designs

MathRelaxing

THE MATHEMATICS

Every knotwork panel in this book is generated by the same rule real Celtic artisans used centuries ago — formalized as a grid algorithm.

The page is divided into a grid of cells. Each cell is randomly assigned one of three tiles: two mirror-image quarter-circle arcs (a Truchet tile), or a straight crossing. Every tile connects all four edge midpoints of its cell, so neighboring tiles always align — no strand ever dead-ends.

Each strand is drawn as a woven ribbon: at every crossing one band passes over and the other is broken to pass under, alternating by grid parity — the rule that defines an alternating knot in knot theory. Three page styles appear: full-bleed woven panels, smaller centered medallions, and mirror-symmetric knots with four-fold symmetry.

Two pages break from the grid to show classic motifs by name: the Trinity Shamrock (a polar rose curve, $r = \cos 1.5\theta$) and the Triskelion (three Archimedean spirals rotated 120° apart).

© MathRelaxing. For personal use only. Not for redistribution or resale.



































































































